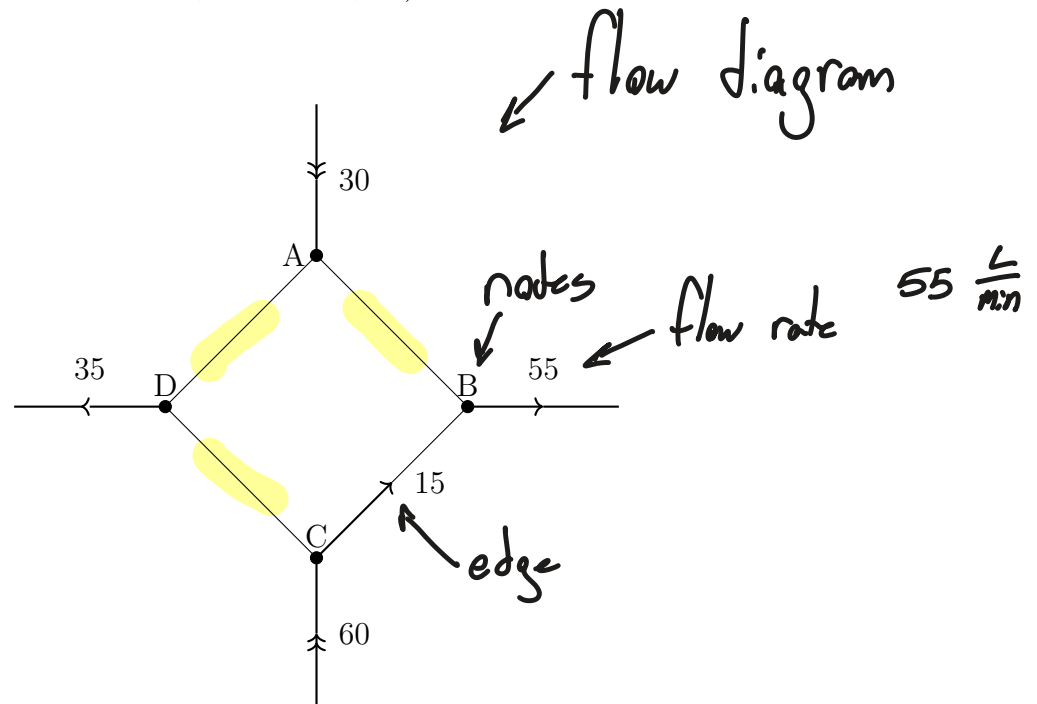
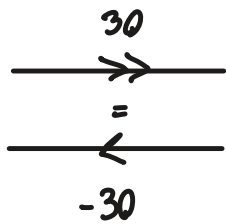


week 1

Lecture 1 (section 1.9)

Applications of linear systems

⇒ Consider a network flow with 4 nodes: A,B,C,D (can represent the **flow** of information, objects — wifi-traffic, water, cars, ...)



⇒ A flow rates into **nodes** are given

⇒ Flow rate along **CB edge** is specified

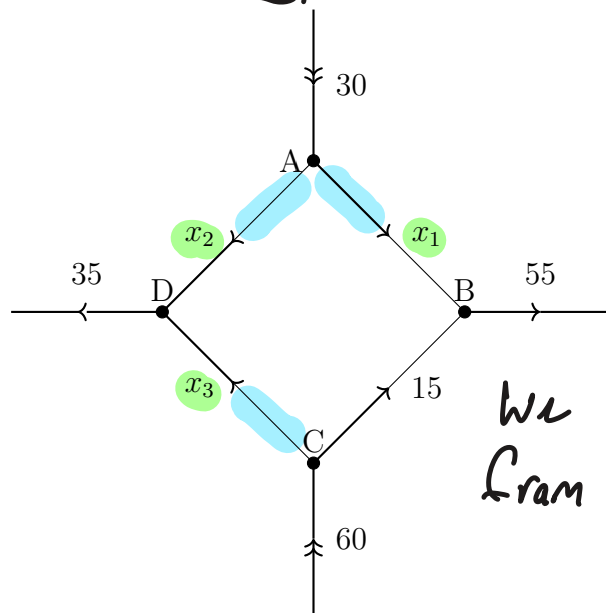
Problem: find the remaining rates through each edge

Basic principle: flow conservation at each node

setting
Equation

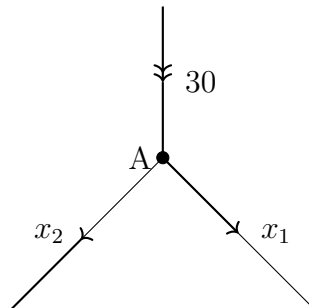
in	out
30	$x_1 + x_2$

- A: $30 = x_1 + x_2$
- B: $x_1 + 15 = 55$
- C: $60 = x_3 + 15$
- D: $x_2 + x_3 = 35$



We assumed flow is from node A to B

• Node A:



$$\underbrace{30}_{\text{"in"}} = \underbrace{x_1 + x_2}_{\text{"out"}}$$

• Node B:

$$x_1 + 15 = 55$$

• Node C:

$$60 = x_3 + 15$$

• Node D:

$$x_2 + x_3 = 35$$

Conservation laws $\implies$ system of linear equations

$$\begin{cases} x_1 + x_2 = 30, & (1) \\ x_1 + 15 = 55, & (2) \\ x_3 + 15 = 60, & (3) \\ x_2 + x_3 = 35, & (4) \end{cases}$$

↑ never forget
use to check

Solve problem $\implies$ solve linear system

Highschool

$$(2): \quad x_1 = 55 - 15 = 40$$

$$(3): \quad x_3 = 60 - 15 = 45$$

$$(1): \quad x_2 = 30 - x_1 = 30 - 40 = -10 \quad (\text{reversed direction})$$

$$(4): \quad \text{DOES NOT CARRY ANY NEW INFORMATION}$$

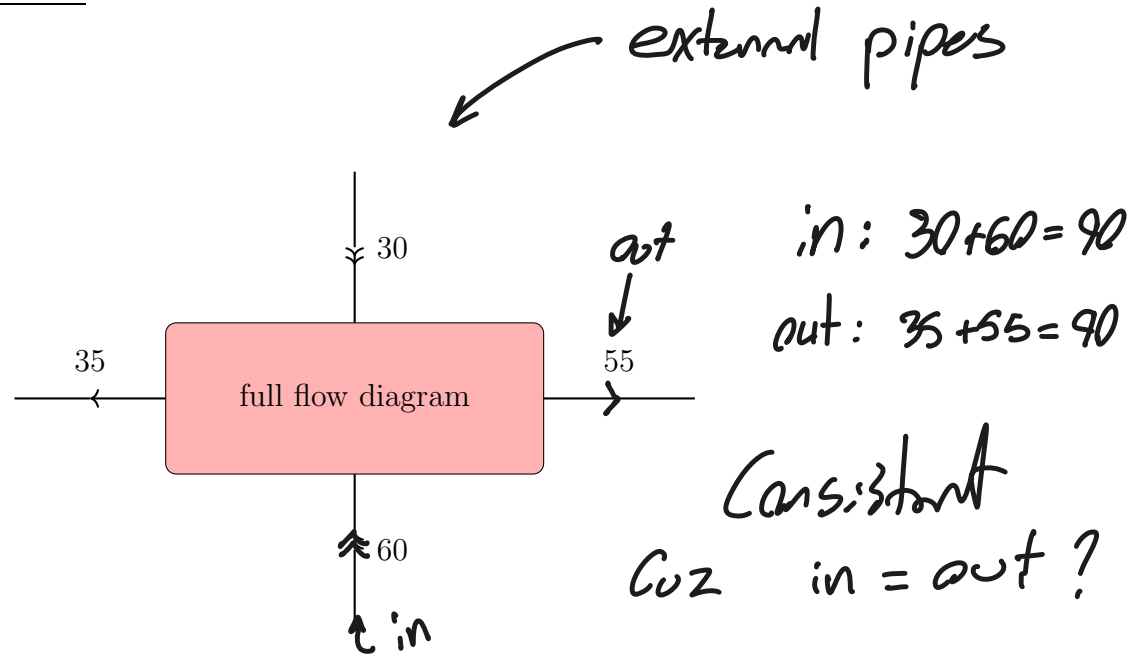
(4) is a *constraint*:

$$x_2 + x_3 = (-10) + 45 = 35 \stackrel{?}{=} 35 \quad (\text{eq.4})$$

$\implies$ Problem is consistent, there is a unique solution

Can have 3
- 1
- 0
- ∞ solutions

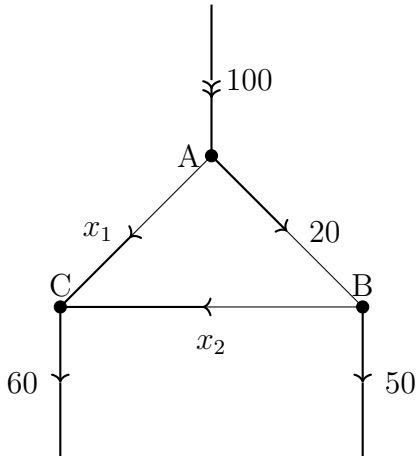
Why consistent?



$\Rightarrow$ Because there is conservation at any node, there must be conservation for the whole network

"IN"		"OUT"
$30 + 60$		$35 + 55$
$\parallel$		$\parallel$
90	$\checkmark$ $\equiv$	90

Solution to linear system does not always exist:



$$\text{node A : } 100 = x_1 + 20 \quad \longrightarrow \quad x_1 = 80$$

$$\text{node B : } 20 = x_2 + 50 \quad \longrightarrow \quad x_2 = -30$$

$$\text{node C : } x_1 + x_2 = 60 \quad \longrightarrow \quad 80 + (-30) = +50 \neq 60$$

Reason:

"IN"

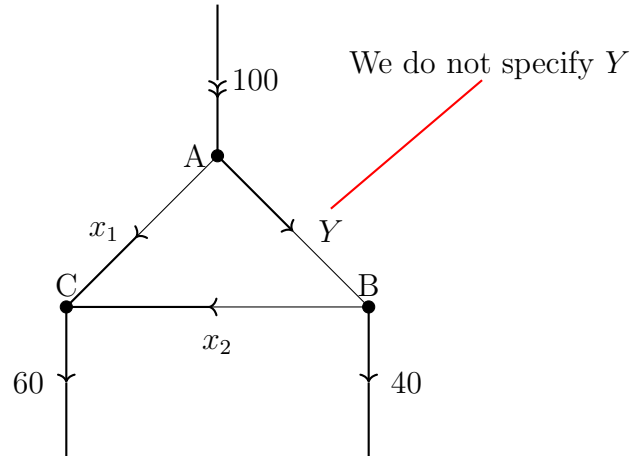
100

$\neq$

"OUT"

$60 + 50 = 110$

Solution to linear system is not always unique:



$$\text{node A : } 100 = x_1 + Y \quad \longrightarrow \quad x_1 = 100 - Y$$

$$\text{node B : } Y = x_2 + 40 \quad \longrightarrow \quad x_2 = Y - 40$$

$$\text{node C : } x_1 + x_2 = (100 - Y) + (Y - 40) = 60 \quad \longrightarrow \quad \checkmark \equiv 60$$

$\implies$ True for any Y

Consistent?

"IN"

$$100$$

=

"OUT"

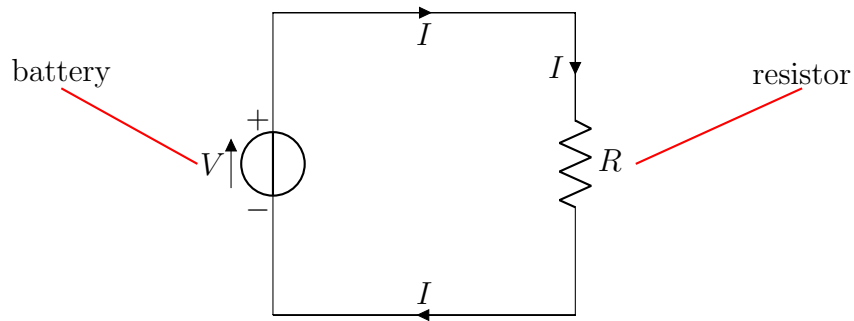
$$60 + 40 = 100$$

$\implies$ In the course we study when the system of linear equations is consistent, and when there is a unique solution

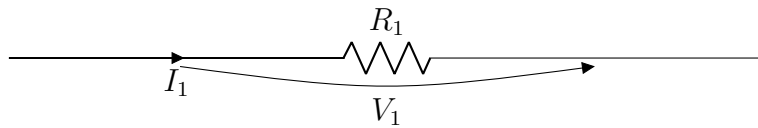
$\implies$ In previous example setting $Y = 20$ gives a unique solution; without specifying Y system of equations is underdetermined

Electric circuits

⇒ Elementary electric circuit:



- **current** I flows from the positive pole of the battery to the negative pole
- battery is characterized by the **voltage** V (electric potential difference between the poles)
- current is characterized by a flow of positively charged particles (convention) — negatively charged particles (electrons) move in the direction opposite to that of the current
- **resistor** R reduces the *electric potential* according to **Ohm's law**:

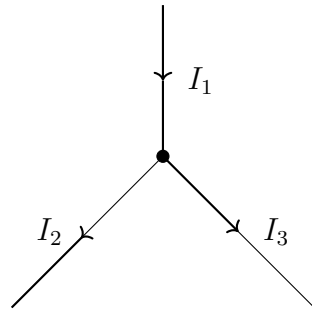


$$V_1 = R_1 \cdot I_1$$

potential drop V_1	resistance of the resistor R_1	current I_1
across the resistor	property of the material	through the resistor
in volts (V)	in Ohms (Ω)	in amperes (A)

Kirchoff's laws of electric circuits:

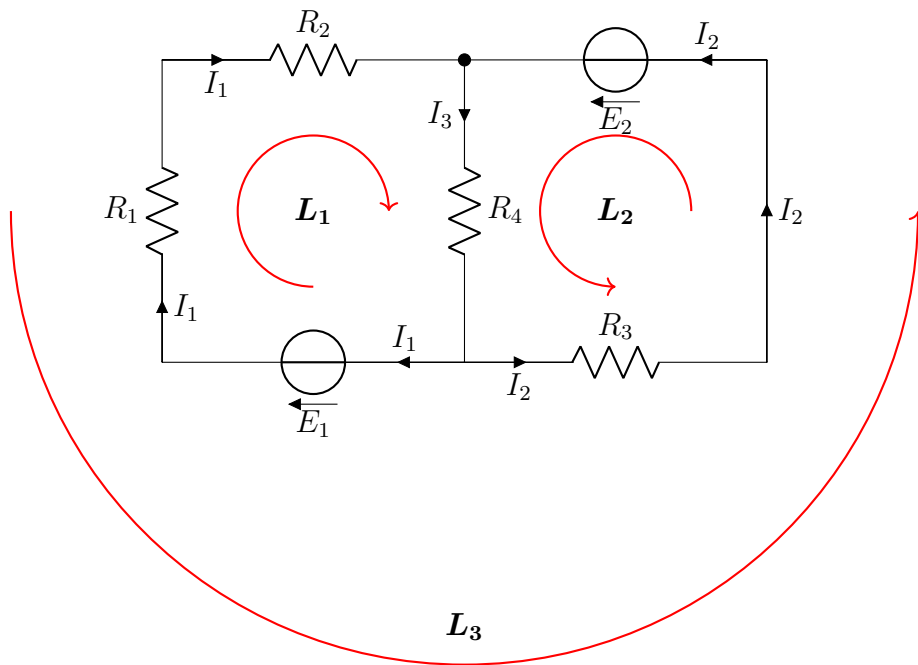
① Current conservation at each node



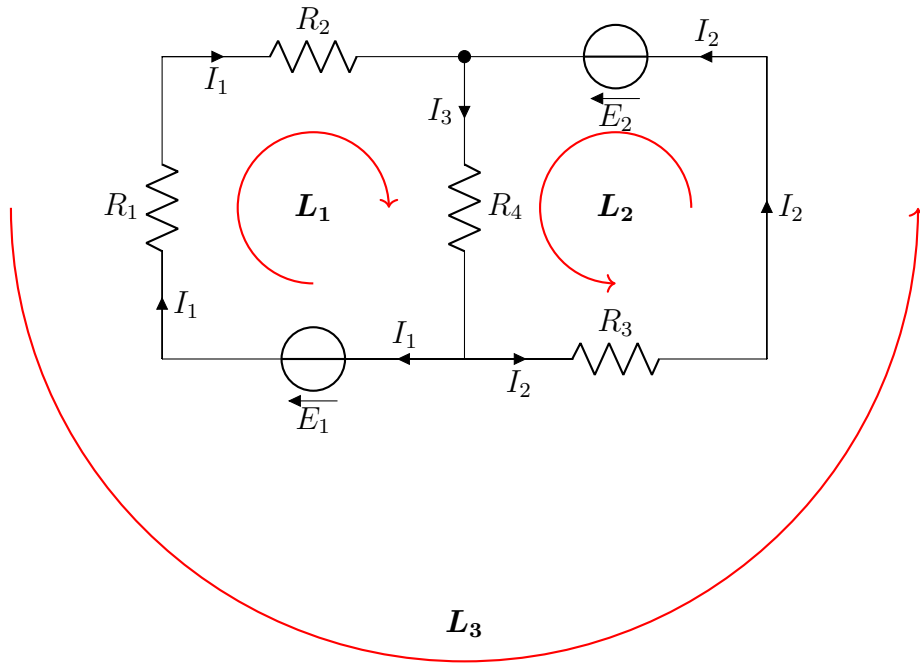
$$\underbrace{I_1}_{\text{"in"}} = \underbrace{I_2 + I_3}_{\text{"out"}}$$

② Sum of voltage rises = sum of voltage drops

$\Rightarrow$ In normal language: electric potential change is zero for any closed loop of the circuit



$\Rightarrow$ There are 3 closed loops: $\{\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3\}$; orientation in red



$$\mathbf{L_1 :} \quad \underbrace{R_1 \cdot I_1 + R_2 \cdot I_1 + R_4 \cdot I_3}_{\text{"voltage drops"}} = \underbrace{E_1}_{\text{"voltage rises"}}$$

$$\mathbf{L_2 :} \quad \underbrace{R_4 \cdot I_3 + R_3 \cdot I_2}_{\text{"voltage drops"}} = \underbrace{E_2}_{\text{"voltage rises"}}$$

$$\mathbf{L_3 :} \quad \underbrace{R_3 \cdot I_2 + (-R_2 \cdot I_1) + (-R_1 \cdot I_1)}_{\text{"voltage drops"}} = \underbrace{(-E_1) + E_2}_{\text{"voltage rises"}}$$

$\Rightarrow$ Note **minus** signs: they appear because we are going 'against' the current direction

$\Rightarrow$ Note:

$$\mathbf{L_3 = L_2 - L_1}$$

that is, loop $\mathbf{L_3}$ is not *independent*